UNIVERSITI TUNKU ABDUL RAHMAN

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MAY 2025 EXAMINATION

UECS3253 WIRELESS APPLICATION DEVELOPMENT

FRIDAY, 30 MAY 2025 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

This question paper consists of **TWO (2)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Section A: Answer **ALL TWO (2)** questions.

Section B: Answer **ONLY ONE (1)** question.

Each question in **Section A** carries 30 marks, question in **Section B** carries 40 marks.

Remark: Should a candidate answer **TWO (2)** questions in **Section B**, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Answer the questions in the answer booklet provided.

UECS3253 WIRELESS APPLICATION DEVELOPMENT

Section A: Answer **ALL** questions.

Q1. (a) Wireless devices are widely adopted in the current world and there are many applications being developed for wireless devices.

(i) Define wireless device and provide **ONE (1)** example of wireless device. (4 marks)

(ii) Consider the following scenario,

> A company is developing a parcel delivery application that integrates WebView to provide the functionalities, e.g., browsing and parcel tracking. There is a mobile push notification to notify the user about the delivery status of the parcel. In addition, the application can access to the device's location services to estimate the distance between the user and parcel collection point.

Identify the suitable type of mobile applications and provide **ONE (1)** justification to support your answer. (4 marks)

(b) Android is one of the most popular mobile operating systems, and it is built on the Linux kernel.

(i) Android is based on the mobile software stack that include multiple layers, and Android runtime (ART) is one of the important layers. Identify **TWO (2)** major features of ART. (4 marks)

(ii) Other than ART and Linux Kernel, provide **ONE (1)** layer in the mobile software stack. (2 marks)

(iii) React Native can be used to develop Android application. Identify the suitable React Native command to create a new project folder namely WAD2, with the React Native version of 0.78. (2 marks)

(c) React Native refers to the JavaScript-based framework and component is one of the important concept of React Native.

(i) Consider the following code segment that imports the React Native component. Identify **ONE (1)** suitable type of export to export the ComponentA and provide the code segment to export the ComponentA.

```
import { ComponentA } from './file'
```

(4 marks)

(ii) Component lifecycle refers to the important concept of class components. Identify **TWO (2)** phases of a React component. (4 marks)

Section A Q1. (c) (Continued)

(iii) Given the code segment that implements functional component, identify the class component-based implementation for line 4.

```
1:   // Assume the necessary Components were
2:   imported
3:
4:   export default function AppB(){
5:     const [a, setA] = useState('0');
6:     return(
7:       <CustomComponent
8:         displayId={a}
9:       />
10:    );
11:  }
```

(2 marks)

(iv) Given the code segment below, identify **ONE (1)** props and **ONE (1)** state.

```
1:   // Assume the necessary Components were
2:   imported
3:
4:   export default function AppB(){
5:     const [a, setA] = useState('0');
6:     return(
7:       <CustomComponent
8:         displayId={a}
9:       />
10:    );
11:  }
```

(4 marks)

[Total : 30 marks]

Section A (Continued)

Q2. (a) Persistent data storage is important in mobile applications as it ensures that user preferences or settings are retained even when the device is restarted. In React Native, SQLite can be implemented for persistent data storage.

(i) Identify the name of the plugin that the react-native-sqlite-storage is created based on. (2 marks)

(ii) Consider the following scenario,

react-native-sqlite-storage supports the execution of SQL queries, including transaction to allow the application to interact with an SQLite database in a React Native application. Consider the developer is developing a React Native application that allows user to purchase stationary. The insertion of the purchasing record is implemented in the way that all the INSERT queries are being executed in the same TRANSACTION. For instance, there are two items in the purchaseItems are supposed to be inserted into the database after the user purchasing the items:

```
const purchaseItems = [
{purchaseId: '0001', name: 'VehicleA', price:
40000},
{purchaseId: '0001', name: 'VehicleB', price:
59900}];
```

The implementation is as follows:

```
db.transaction(tx => {
    purchaseItems.forEach( item => {
        tx.executeSql(      'INSERT      INTO
purchaseItem (purchaseId, name, price) VALUES
(?, ?, ?);',
        [item.purchaseId,           item.name,
item.price],
            callback
        )
    })
})
```

Analyse the scenario above and provide justification on implementing the INSERT queries in the same TRANSACTION. (4 marks)

Section A Q2. (a) (Continued)

(iii) By referring to the code segment below, construct the complete code to perform SQL query on the table vehicleInfo, so that the record in the vehicleInfo is changed from **Table 2.1** to **Table 2.2**.

Table 2.1: vehicleInfo

vehicleId	vehicleModel	vehiclePrice
1	VehicleA	59900
2	VehicleB	59900

Table 2.2: vehicleInfo after performing SQL query

vehicleId	vehicleModel	vehiclePrice
2	VehicleB	59900

```
const methodA = async (
   db: SQLiteDatabase,
) => {
// Complete the code here
const query =

try {
    // Complete the code here

  } catch (error) {
    ToastAndroid.show("Failed");
    throw Error("Failed")
  }
}
```

(4 marks)

(b) AsyncStorage refers to the storage method that leverages key-value pair to store the data.

(i) Suppose the package for AsyncStorage is not installed in the React Native project folder, construct the command to install the package.

(2 marks)

(ii) By referring to the code segment below, construct the code to remove the information that was stored using AsyncStorage. The deletion should be done based on the keys in the vehicleInfoKey, and the function multiRemove should not be used.

Section A Q2. (b) (ii) (Continued)

```
// Assume the code is in an async function

const vehicleInfoKey = ['model', 'price'];

try {
    for (const key of vehicleInfoKey) {
        // Complete the code here
    }
} catch (error) {
    console.log('Error occurs: ', error);
}
```

(4 marks)

(iii) In the code segment below, the data from AsyncStorage supposed to be displayed to the user through the state variable a. However, there is an incorrect implementation that causes the application is not displaying the data correctly. Suggest a solution to overcome this situation.

```
// Assume necessary components/functions were
imported

const App = () => {
    const [a, setA] = useState('0');
    useEffect(()=>{
        _readA();
    },[]);

    const _readA = async () => {
        try {
          const value = await
AsyncStorage.getItem('a');
        } catch (error) {
          console.log('Error reading value:',
error);
        }}

    return(
        <View>
            <Text>{a}</Text>
        </View>
    )
};

export default App;
```

(4 marks)

Section A Q2. (Continued)

(c) Location services in a mobile application are useful for applications, such as maps or location-based notifications.

(i) It is required to modify the manifest file, that is, AndroidManifest.xml so that the application can obtain the location information. Construct the required location permission in the <uses-permission /> tag for getting approximate network-based location. (2 marks)

(ii) By referring to the code segment below, construct the code segment that allows createMapLink() to obtain the route of public transport between locationA to locationB. The information is obtained through apple.

```
// Assume the necessary Components were
imported
const data = {
    // Complete the code here
};

const getLink = () => {
     return createMapLink(data);
}

const App = () => {
return(
    <>
    <WebView
        source={{ uri: getLink() }}
        style={{
           width:deviceWidth,
           height:deviceHeight}}
           scalesPageToFit={true}
         />
    </>
);
}

export default App;
```

(8 marks)

[Total : 30 marks]

Section B: Answer ONLY **ONE (1)** question.

Q3. Connection between Server application and React Native application is important.

(a) Consider the scenario that your company is developing an attendance tracking application that integrates with third-party HR software using APIs. The HR software provides a RESTful API for retrieving employee attendance records.

(i) Define an Application Programming Interface (API) and identify **TWO (2)** common uses of APIs in software development. (5 marks)

(ii) Identify **TWO (2)** differences between API and web service and provide **TWO (2)** examples of an API that is not a web service. (4 marks)

(iii) Consider the server could respond to the user with **THREE (3)** possible HTTP status codes including 201, 400 and 403, explain the meaning of the status codes in this context. (6 marks)

(b) Your company has implemented the Employee Retrieve functionality for making HTTP requests in its React Native application. Initially, the company implemented the Employee Update functionality using Axios as shown in the code segment below. However, due to network compatibility issues on certain devices, the company has decided to switch to the Fetch API.

```
import axios from "axios";
import { ToastAndroid, Alert } from "react-native";
import { useNavigation } from "@react-
navigation/native";

const _updateEmployee = async () => {
    const navigation = useNavigation();
    let url = "https://hrapp.com/api/updateemployee";

    try {
        const response = await axios.post(url, {
            empId: "1234",
            salaryAmount: "3000"
        });

        if (response.success) {                    //
            ToastAndroid.display("Updated",
ToastAndroid.SHORT);
            navigation.go("EmployeeDetails", { id:
response.updateID }); //
        } else {
            Alert.alert("Update Failed");
        }
    } catch (error) {
        console.log(error); //
    }
};
```

Section B Q3. (b) (Continued)

Your task is to rewrite the existing Axios implementation using Fetch while ensuring the following requirements are met:

Requirements:

- The application should send an **HTTP PUT** request to update employee data at: **https://hrapp.com/api/updateemployee**.
- The data should be formatted as **JSON** with the following fields:
  - employeeId: "10"
  - salary: "5000"
- If the update is **successful**, the application should show a **Toast message** and **navigate** to the "EmployeeDetails" screen, passing updateID as a route parameter.
- If the update **fails**, the application should display an **Alert message**.

(15 marks)

(c) The following code segment shows the implementation of the Employee Retrieve function using Axios, which contains errors.

```
import axios from "axios";
import { ToastAndroid, Alert } from "react-native";
import    {    useNavigation    }    from    "@react-
navigation/native";
import { useState } from "react";


const [employee, setEmployee] = useState({});

const _retrieveEmployee = async () =>
{
const navigation = useNavigation();

let url="https://hrapp.com/api/retrieveemployee";

     try {
const response = await axios.post(url);

ToastAndroid.display (
"Employee Found", ToastAndroid.SHORT
);

setEmployee(response.empData);
}

       else   {   Alert.alert("Retrieval   Failed");
navigation.navigateBack();
}
}
catch (error) { console.log(error); } };
```

Section B Q3. (c) (Continued)

Your task is to rewrite it using Fetch while ensuring the following are correctly handled:

- Use **HTTP GET** as the HTTP method.
- Convert the response to **JSON**.
- If successful, show a **Toast message** and set the value to employee.
- If unsuccessful, display an **Alert message** and navigate back to the previous screen.
- Handle any **network errors** properly. (10 marks)

[Total : 40 marks]

Q4. WebSocket is another approach that allows React Native to connect to server application.

(a) Answer the following questions related to Socket.IO.

(i) By referring to the code segment below, identify the correction configuration so that the Socket.IO will only use the WebSocket protocol.

```
var socket =
io('http://193.188.1.26:8888/voiceAPI', {
    // Complete the code here
});
```

(1 mark)

(ii) Define Socket.IO and explain **TWO (2)** advantages of using Socket.IO. (5 marks)

(iii) Analyse the code segment below and identify whether the **THREE (3)** socket instances are sharing the same manager instance. Provide **ONE (1)** justification to support your answer.

```
var socket = io("http://localhost.com");
var socketOne = io("http://localhost.com/one",
{ multiplex: false });
var socketTwo = io("http://localhost.com/two" ,
{ multiplex: false });
```

(4 marks)

Section B Q4. (a) (Continued)

(iv) Construct the code segment to create **THREE (3)** socket instances that are not sharing the same manager instance. The requirements are as follows:

- The first socket instance connects to https://wadfe.com.
- The second socket instance connects to https://wadfe.com/voice.
- The third socket instance connects to https://wadfe.com/chat.
- forceNew option is used. (6 marks)

(b) Consider the company is developing a React Native application that using biometrics API to verify the user and receive updates related to the verification process. Briefly, the biometrics API can be connected using Socket.IO. Complete the code segment below based on the requirement in the Q4 (b) (i – iv), so that the React Native application can communicate with the biometrics API.

(i) Consider the current implementation of var socket is incorrect, construct the correct implementation based on the following requirements:

- The web server is hosted with Transport Layer Security (TLS) supported.
- Other than polling, WebSocket is also used as the communication method. (6 marks)

(ii) Upon connected to the server, the React Native application display a console message "connected to server" and reset the biometric data that is stored in the state variable. (6 marks)

(iii) Upon receiving the event "matching_result", the handler first converts the data from JSON string to JSON object. After that, the React Native application navigates to the screen namely "screen" by passing the matching result through route parameter. (6 marks)

(iv) The React Native application submit username and biometric data to the server. (6 marks)

```
// Assume the necessary components/functions
were imported

/*
  Complete the code here for Q4 (b) (i)
*/
var socket =
io.establish('http://www.FE2025.com/biometrics
API', {
    transports: ['polling'],
});
const App = ({route, navigation}: any) => {
```

Section B Q4. (b) (iv) (Continued)

```
Const [biometricTemplate,
setBiometricTemplate] = useState({});
const [username, setUsername] = useState('');
const  [mathingResult,  setMatchingResult]  =
useState('');

useEffect(()=>{
    // A new handler is registered
    /*
        Complete the code here for Q4 (b) (ii)
    */


    // A new handler is registered
    /*
        Complete the code here for Q4 (b) (iii)
    */
},[]);

return(
    <ScrollView>
            <CustomBiometricExtractor
                onCapture = { (template) => {

setBiometricTemplate(template)
                }}
            />
            <TextInput
                value = { username }
                onChangeText = { (input: any)
=> {
                        setUsername(input);
                } }
            />
            <Button
                title = 'Verify User'
                onPress = {()=>{
           /*
        Complete the code here for Q4 (b) (iv)
           */
                }}
            />
    </ScrollView>
);
}
export default App;
```

[Total : 40 marks]